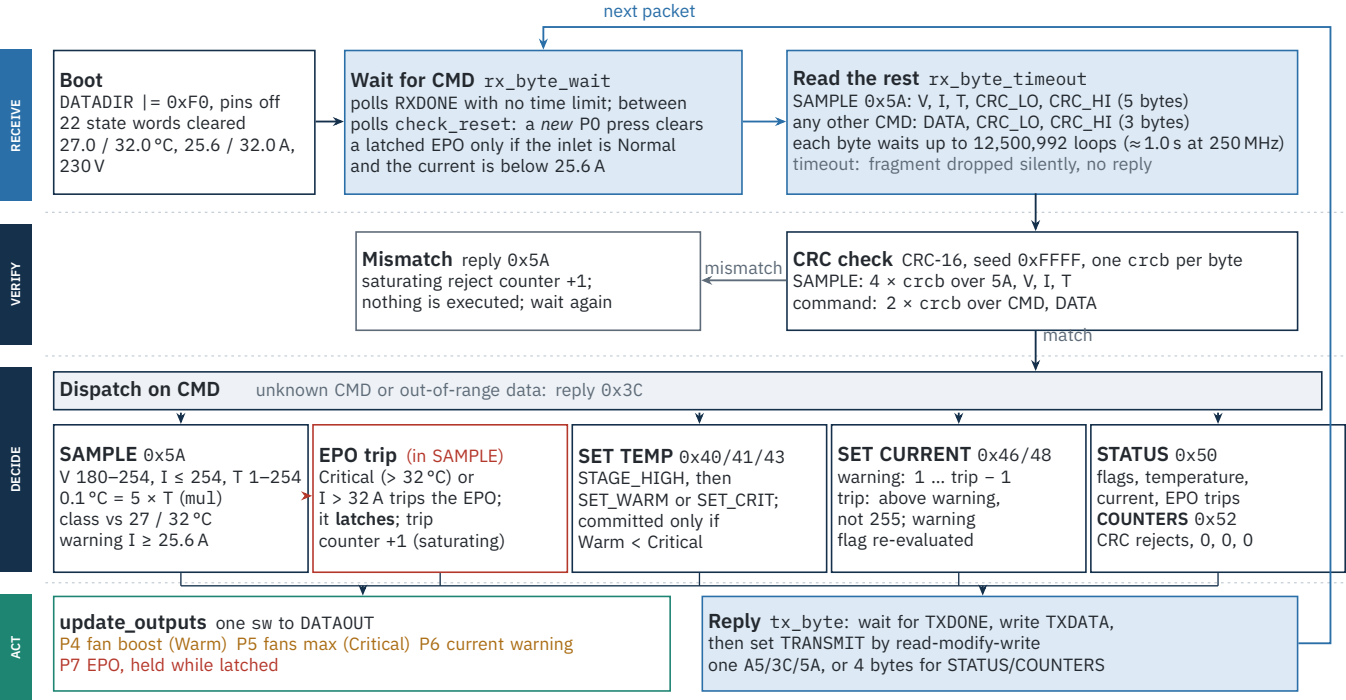
next packet
RECEIVE
Boot
DATADIR |= 0xF0, pins off
22 state words cleared
27.0 / 32.0 °C, 25.6 / 32.0 A, 230 V
Wait for CMD rx_byte_wait
polls RXDONE with no time limit; between polls check_reset: a new P0 press clears a latched EPO only if the inlet is Normal and the current is below 25.6 A
Read the rest rx_byte_timeout
SAMPLE 0x5A: V, I, T, CRC_LO, CRC_HI (5 bytes)
any other CMD: DATA, CRC_LO, CRC_HI (3 bytes)
each byte waits up to 12,500,992 loops (≈ 1.0 s at 250 MHz)
timeout: fragment dropped silently, no reply
VERIFY
Mismatch reply 0x5A
saturating reject counter +1;
nothing is executed; wait again
mismatch
CRC check CRC-16, seed 0xFFFF, one crcb per byte
SAMPLE: 4 × crcb over 5A, V, I, T
command: 2 × crcb over CMD, DATA
match
DECIDE
Dispatch on CMD
unknown CMD or out-of-range data: reply 0x3C
SAMPLE 0x5A
V 180–254, I ≤ 254, T 1–254
0.1 °C = 5 × T (mul)
class vs 27 / 32 °C
warning I ≥ 25.6 A
EPO trip (in SAMPLE)
Critical (> 32 °C) or I > 32 A trips the EPO;
it latches; trip counter +1 (saturating)
SET TEMP 0x40/41/43
STAGE_HIGH, then SET_WARM or SET_CRIT;
committed only if Warm < Critical
SET CURRENT 0x46/48
warning: 1 ... trip − 1
trip: above warning, not 255; warning flag re-evaluated
STATUS 0x50
flags, temperature, current, EPO trips
COUNTERS 0x52
CRC rejects, 0, 0, 0
ACT
update_outputs one sw to DATAOUT
P4 fan boost (Warm) P5 fans max (Critical) P6 current warning
P7 EPO, held while latched
Reply tx_byte: wait for TXDONE, write TXDATA,
then set TRANSMIT by read-modify-write
one A5/3C/5A, or 4 bytes for STATUS/COUNTERS